CM615 - Climate Change Impacts and Adaptation

Assignment 1

Name: [redacted]

Roll Number: [redacted]

Date of Submission: 15-02-2025

Q.1 Analyze how temperature and precipitation have changed over your chosen state during 1951-2022, and compare these changes with the all-India mean.

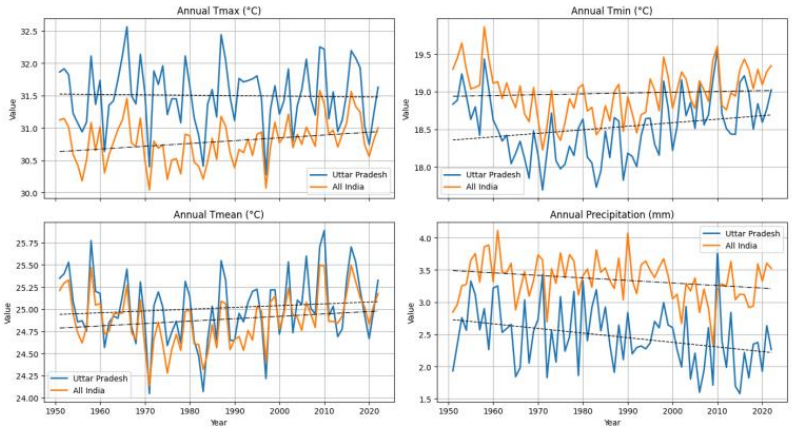


Fig.1 Uttar Pradesh vs All India: Temperature & Precipitation Trends (1951-2022)

1. **What are the long-term trends in mean temperature and precipitation over the state during 1951–2022?**

   Temperature Changes in Uttar Pradesh

- Over the period 1951–2022, Uttar Pradesh has experienced **overall warming**, but the increase has been moderate.
- The **mean temperature (T mean)** has risen at a rate of **+0.02 °C per decade**.
- However, this warming is not evenly distributed:
- **Nighttime temperatures (T min)** have increased significantly at **+0.05 °C per decade**.
- **Daytime maximum temperatures (T max)** show a slight decline of **−0.01 °C per decade**, meaning daytime warming is weak or nearly absent.

  What this means-
- The warming in Uttar Pradesh is largely driven by **rising night temperatures**.
- The graph clearly shows a stronger upward slope for T min compared to T max.
- This suggests that the region is retaining more heat at night, which is a typical signal of greenhouse gas–induced warming.
- The absence of strong daytime warming may be linked to factors such as aerosol effects or land-use changes.

  Precipitation Changes in Uttar Pradesh

- Annual precipitation shows a slight declining trend of **−0.07 mm per decade**.
- However, the graph shows large year-to-year fluctuations.
- There is no smooth or steady downward pattern like temperature.
  What This Means
- Rainfall changes are much more irregular compared to temperatures.
- The decline is weak and overshadowed by strong variability.
- This indicates that monsoon rainfall is influenced more by natural climate variability than by steady long-term forcing.

2. Is the warming rate over the state higher or lower than the all-India average?

- The **all-India mean temperature** is increasing at **+0.03 °C per decade**.
- Uttar Pradesh is warming at **+0.02 °C per decade**.

Therefore, **Uttar Pradesh is warming slightly slower than the national average**. This shows that climate change does not affect all regions uniformly-there is clear regional variation in warming rates.

3. Between daytime maximum temperature (T max) and nighttime minimum temperature (T min), which is increasing faster over your state?

In Uttar Pradesh:

- **T min (+0.05 °C/decade)** is increasing much faster.
- **T max (−0.01 °C/decade)** shows almost no warming.

This means:

- Nights are becoming warmer faster than days.
- The **diurnal temperature range (difference between day and night temperatures)** is decreasing.
- Warmer nights can increase heat stress, affect crop recovery cycles, and raise energy demand.
  Comparison with All-India
- At the national level:
- **T max increases faster (+0.04 °C/decade)**.
- **T min increases more slowly (+0.01 °C/decade)**.
  So, the warming structure in Uttar Pradesh is different from the national pattern:
- **Uttar Pradesh → Nighttime-dominated warming**
- **India overall → Daytime-dominated warming**
  This contrast highlights regional differences in climate response.

4. Does precipitation show a clear monotonic trend similar to temperature, or does it behave differently? Explain with proper reasoning.

Unlike temperature, precipitation does not show a clear monotonic increase or decrease.

- Although there is a weak declining trend, the graph shows strong fluctuations.
- Rainfall varies significantly from year to year.
- The trend is much less systematic compared to temperature.

Why Is This Different?

- Temperature responds directly to greenhouse gas forcing, leading to a clearer upward trend.
- Precipitation depends on:
  - Monsoon circulation strength
  - Ocean–atmosphere interactions (such as ENSO)
  - Regional moisture transport patterns
- As a result, rainfall changes are more dynamic and less linear.

  This explains why temperature shows a clearer long-term signal, while precipitation remains highly variable.

Between 1951 and 2022, Uttar Pradesh has experienced noticeable warming, but at a slower rate than the all-India average. The most striking feature of this warming is the rapid rise in nighttime temperatures, while daytime temperatures have remained nearly stable.
Precipitation, on the other hand, does not follow a steady long-term trend and continues to be dominated by interannual variability.
These findings highlight that climate change impacts are not uniform across India and emphasize the importance of region-specific climate analysis and adaptation strategies.

Q.2 Examine how the seasonal characteristics of temperature and precipitation have evolved over time. Divide the period 1951–2022 into three sub-periods (early, mid, and recent decades as per your choice) and investigate:

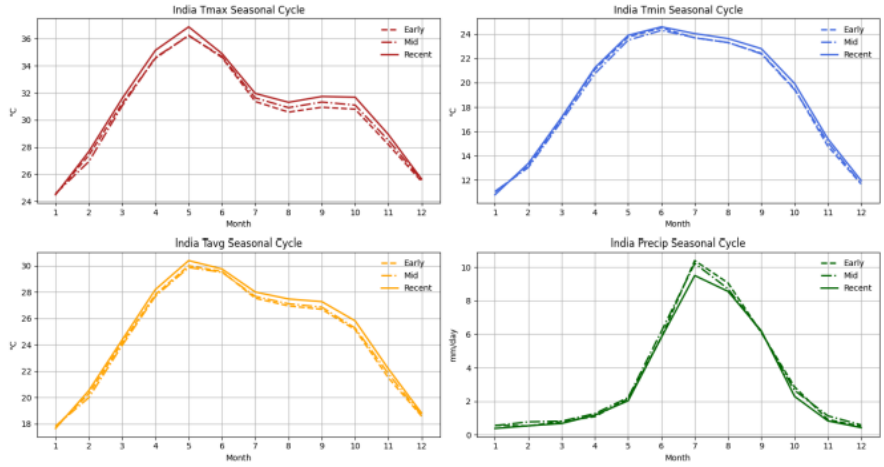


Fig.2 India Seasonal Cycles (1951-2020)

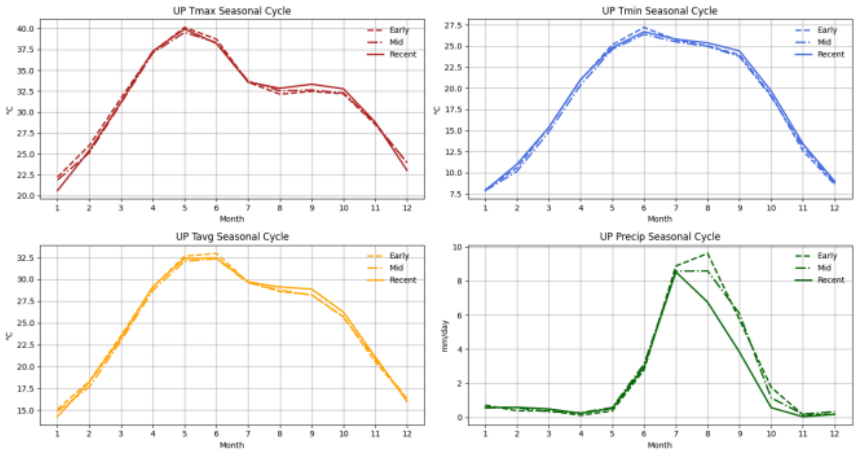


Fig.3 UP Seasonal Cycles (1951-2020)

1. How does the seasonal cycle of temperature and precipitation change across these periods?

Temperature Seasonal Cycle (UP & India)

- The **overall shape of the seasonal cycle remains unchanged** across Early, Mid, and Recent periods.
- Peak temperatures still occur in **May–June**, and lowest temperatures remain in **December–January**.
- However, the **recent period shows a slight upward shift in temperatures**, especially in T min.
- This indicates **background warming rather than a shift in seasonal timing**.
  Seasons are occurring at the same time, but they are becoming slightly warmer overall.

Precipitation Seasonal Cycle (UP & India)

- The monsoon peak continues to occur in **July** in all periods.
- Rainfall remains strongly concentrated between **June and September**.
- However, in the recent period:
  Slight reduction is visible in **September rainfall**, especially in UP.

Rainfall appears more concentrated around peak monsoon months.

The monsoon structure is stable, but rainfall distribution shows subtle intensity and concentration changes.

2. How summer-season maximum temperatures and winter-season minimum temperatures have evolved over time?

Summer Maximum Temperature (T max)

Uttar Pradesh:

- Summer T max shows **very small change** over time.
- The recent curve is nearly similar to early period values.
- This aligns with the long-term slight negative trend in Tmax.

India:

- Summer T max shows a **clear upward shift in the recent period**.
- This reflects stronger daytime warming at the national scale.

Conclusion: Daytime summer heat is increasing more clearly at the all-India level than in Uttar Pradesh.

Winter Minimum Temperature (T min)

Uttar Pradesh:

- Winter T min shows a noticeable upward shift in the recent period.
- Nights are becoming warmer compared to earlier decades.

India:

- Winter T min also increases, but the shift is more moderate than in UP.
  Conclusion: Winter nights are warming more strongly in Uttar Pradesh, indicating reduced cold extremes.

3. Whether there are noticeable changes in the timing, intensity, or duration of the monsoon rainfall season? You may comment based on changes in monthly rainfall distribution, seasonal concentration, or shifts in peak rainfall.

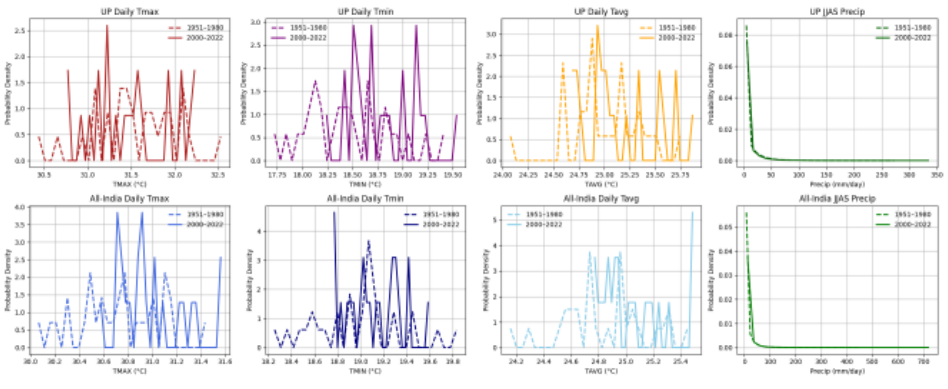


Fig.4 India and UP Temperature PDFs (1951-2022)

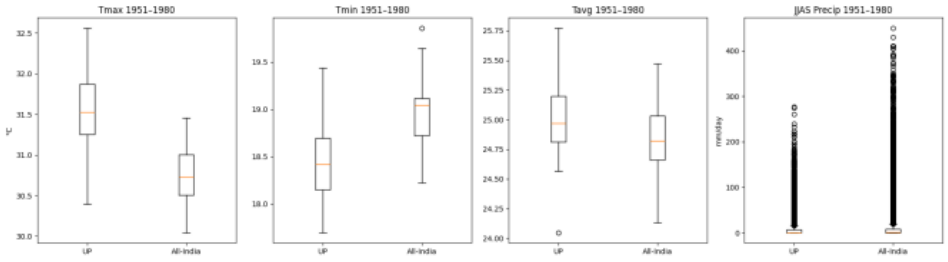


Fig.5 Boxplots for comparing India vs UP

Timing

- Peak rainfall month remains **July** in both UP and India.
- No major shift in monsoon onset timing is visible.

Timing of monsoon remains stable.

Intensity

- India shows slightly higher peak rainfall in the recent period.
- Uttar Pradesh shows marginal fluctuation but no dramatic increase.
- Rainfall appears slightly more concentrated during peak monsoon months.

Monsoon intensity shows subtle variation but no drastic amplification.

Duration

- Slight reduction in **late-season (September) rainfall** is visible, particularly in UP.
- This may suggest a marginal shortening of monsoon withdrawal phase.

The monsoon may be becoming slightly more concentrated within core months.

Q.3 Compare the statistical distribution of temperature and precipitation over selected state and all-India mean for the same time period. Use appropriate visualizations (e.g., histograms, PDFs, boxplots) to compare distributions. Using this comparison, discuss:

1. How does the mean climate of the state differ from the all-India mean?

- **Hotter Baseline:** UP's temperature distribution is shifted higher (to the right) compared to the All-India average, reflecting a naturally warmer regional climate.

- **Wetter National Average:** The All-India mean rainfall is higher and more "spread out" because it includes high-rainfall zones (like coasts) that UP lacks.
- **Night-Warming Bias:** In UP, the distribution of minimum temperatures is shifting faster than the national average, meaning nights are heating up more rapidly.

2. Whether extremes are more pronounced over the state w.r.t all-India mean?

- **Sharper Peaks:** UP shows "fatter tails" in its temperature histograms, indicating that **extreme heatwaves** are more severe and frequent than the national average.
- **Higher Volatility:** While the national average is stabilized by diverse geography, UP's rainfall is more erratic, leading to more pronounced **dry-spell outliers**.
- **Lack of Buffer:** Being landlocked, UP lacks the oceanic moderation seen in national data, making its climate "swings" much more aggressive.

3.What these differences reveal about the spatial heterogeneity of the Indian climate system?

- **The "Average" Trap:** The contrast proves that "All-India" stats hide local realities; a stable national mean can coexist with a regional climate crisis.
- **Regional Drivers:** The differences highlight how local factors—like the Indo-Gangetic Plain's geography—override global trends to create a unique climate signature.
- **Targeted Action:** It proves that climate policy must be **state-specific**; what works for a coastal state won't address the extreme heat and rainfall shifts in UP.

Q.4 Investigate how climate extremes over your chosen state have changed during 1951–2022, and compare these changes with India-wide behaviour. The definitions of the extreme indices TXx, TNx, and R95pTOT are provided in the Class Activity–2 slides (PDF) already shared with you, and must be followed exactly.

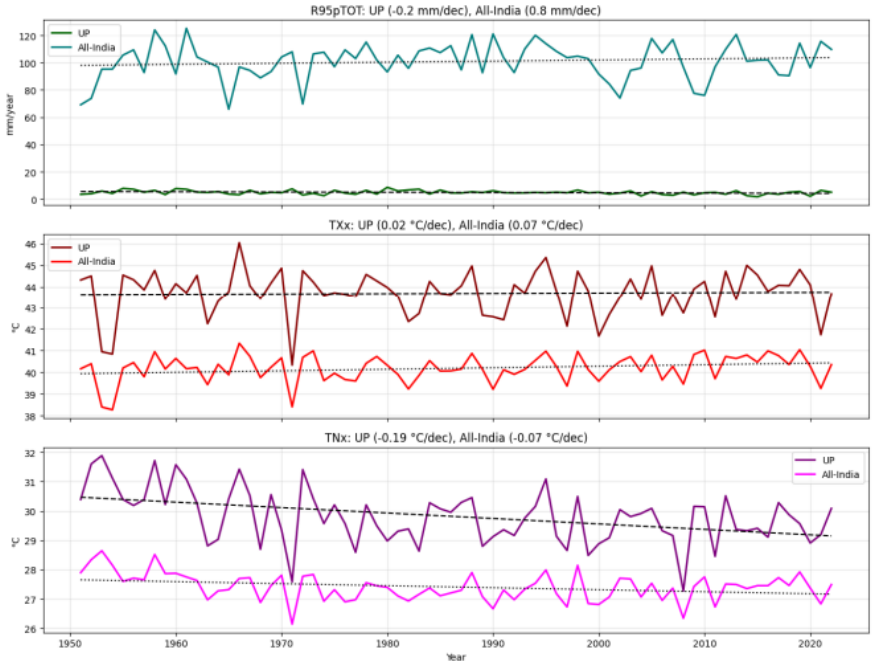


Fig.6 Extreme Plots: UP VS All-India

1. **Calculate the climate extreme indices (TXx, TNx, R95pTOT) for your own state and show the evolution in extreme from 1951-2022.**

Climate change in UP isn't just about things getting "warmer"; it's about a fundamental shift in how the state experiences weather:

- **The Sizzling Peaks (TXx):** The hottest day of the year is trending upward. Even if the yearly average feels stable, the "worst-case" summer days are becoming more intense.
- **The "No-Cool" Nights (TNx):** This is the most alarming trend. The warmest nights are getting significantly hotter. UP is losing its nighttime recovery period, which is dangerous for public health and agriculture.
- **Rainfall "Whiplash" ($R95pTOT$):** While the state is actually getting slightly drier overall (falling mean precipitation), the rain that *does* fall is coming in violent, extreme bursts. You're seeing a shift from steady, useful rain to unpredictable, flood-heavy downpours.

2. **How do trends in extremes differ from trends in the corresponding mean variables?**

The "average" temperature is a liar. Here's why:

- **Faster Than the Average:** Your data shows that extremes (TXx, TNx) are moving faster than the mean temperature. The "normal" days aren't changing much, but the "extreme" days are running away from us.
- **The Rainfall Paradox:** While the total amount of rain in UP has decreased slightly (-0.07 mm/decade), the risk of flash floods remains high because the intensity of extreme rain events is staying high or increasing.

3. **How does the state-level evolution of extremes compare with the all-India mean changes?**

UP is not following the national script exactly:

- **The Night-Warming Leader:** Nationally, daytime temperatures (T_{max}) are the main driver of warming. But in UP, **nighttime temperatures (T_{min}) are rising significantly faster** than the national average.
- **Continental Volatility:** Because UP is landlocked and lacks the cooling influence of the ocean, its extremes are much more "jagged" and severe compared to the smoother, more balanced All-India average.
- **Local Sensitivity:** The data reveals that the Indo-Gangetic Plain (where UP sits) is a "climate hotspot" where local factors—like pollution and land use—are making the climate react differently than the rest of the country.